COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

LEARNER	ARTIFACT	STATUS
Luis Barajas	Problem Brief	Working document

SECTION 01

Visible Failure

The MR-17 code was being run despite many attempts to administer a stop request.

SECTION 02

Verified Facts

According to the scheduler, the record that shows what assignments that MR-17 was and will be given, MR-17 was not given any previous assignments before K-44, which does not appear in its catalogue of tests.
Additionally, the maintenance record confirms that K-44 was removed from regular testing before the incident.
Finally, Nia's incident log shows that MR-17 was given the K-44 command on repeat, with the stop request being unable to stop K-44, citing "ACTIVE_INSTRUCTION_PENDING."

SECTION 03

Assumptions and Open Questions

Nia reported that MR-17 was working as normal before the repeated command. To affirm this account, I would need a record of the commands that MR-17 was given before the time of the incident, as well as its intended instructions, in order to confirm.

The maintenance record recounts that K-44, the administered command, was retired 186 days before, but does not establish that K-44 was completely retired from every service, just from MR-17.

SECTION 04

System Environment Map

SECTION 04A

People and Roles

Nia Voss supervises the QA, acting as support when it does not fulfill standard procedures.
Elias Chen manages the importation of MR-17's commands as the Robotics Systems Engineer.
Mara Ilyan manages the operations of the system, and is the most affected by it.

SECTION 04B

Processes and Rules

The incident log shows the timeline of events that occurred when MR-17 was administered the faulty command.

SECTION 04C

Information and Data

The scheduler shows the previous and future assignments meant for MR-17, which is valuable to understand when the problem started.
The maintenance record shows different codes and commands that were removed, such as K-44, or added, but does not confirm whether or not that every incarnation of the command was removed.

SECTION 04D

Technology and Tools

The command given to MR-17 was in the exact format that the system accepts, and originated from one of the internal interfaces with "legacy_sync.exe" as the source label. The service, as well as who is running it, remains unverified.

SECTION 04E

Environment and Constraints

Paused production in the QA cell prevents production, and thus, profits.

SECTION 05

Investigation Boundary

Inside the Investigation

The team will now examine the source of the faulty command, as well as debate whether or not to restart MR-17 to potentially clear the faulty command.

Outside for Now

The services that still have a copy of the K-44 are unimportant now, but can be resolved later. Figuring out a way of stopping the command or pinpointing the source of the command to terminate it is much more valuable, as it will allow production to resume normally.

SECTION 06

Organizational Impact

According to Mara, the QA cell is being briefly halted, preventing regularly scheduled tests from going off.

SECTION 07

Problem Statement

Production is being halted in QA due to an unforseen administration of the command K-44, which had since been discontinued. This problem affects all of the Stakeholders, primarily Nia, Elias, and Mara, who manage the system. Without understanding either the source or cause of the command, or the effects of restarting MR-17, production will be halted until MR-17 is properly fixed.

SECTION 08

Next Investigation Step

Figuring out the source of the faulty command, K-44, would allow us to figure out how and where this command is being sent, and potentially allow us to figure out a way to prevent it.

SHARED ARTIFACT

Working Document Link

Not entered

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.